

Task

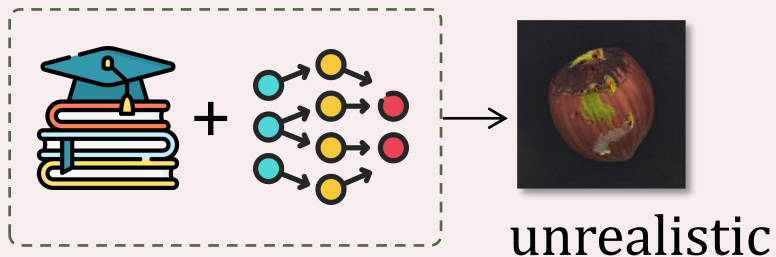
Normal
Image



Based on the provided image (Category: **hazelnut**) to synthesize a high-quality and physically realistic “**hole**” anomaly.

Traditional Methods

Zero-shot (e.g. CutPaste, AnomalyAny)



Knowledge-driven

✗ Low semantic realism

Few-shot (e.g. DFMGAN, AnoDiff)

Training Data



✗ Limited Defect Generalization

AnomalyAgent(Ours)



Environment

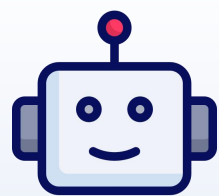
Prompt Generation

Image Generation

Quality Evaluation

Knowledge Retrieval

Mask Generation



Tool Call

Feedback

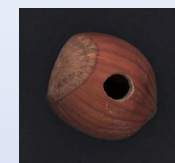
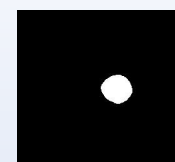
✓ Open-set Generalization
✓ High semantic fidelity

Thought

Action

Observation

Complete Task



Anomaly
Image
& Mask

Single-step Generation

✗ No reasoning

✗ No reflection

✗ No iterative improvement

Closed-loop Synthesis

✓ Structured reasoning

✓ Self-reflection

✓ Iterative refinement